MASTER'S THESIS DEFENSE

Factors Influencing Consumer Purchase Intention in the Singapore Online Grocery Market

A Quantitative Study

Technology Acceptance Model (TAM) Framework Application

Presented by

Nguyen Thi Nhu Nguyen

Master of Science in International Business Management

Supervisor

Philip Snelle

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Introduction

Research Problem & Objectives

Research Gap

Limited academic literature examining online grocery purchase intention specifically in **Singapore's unique market context**, despite the country's advanced digital infrastructure.

Problem Statement

Despite rapid global e-commerce growth, Singapore's online grocery market remains **underdeveloped (only 3% of total consumption)** due to strong traditional retail infrastructure and cultural shopping habits.

Research Objectives

1. **Review** global online grocery market trends and factors influencing consumer purchase intention
2. **Identify** key determinants and analyze how they shape consumer action and behavior patterns
3. **Analyze** the effectiveness of identified factors toward consumer purchase intention
4. **Recommend** strategic improvements to enhance online grocery purchasing adoption

Research Aim

To examine factors affecting consumer purchase intention in Singapore's online grocery sector and quantify their degree of influence on consumer decision-making.

Market Context

Singapore Online Grocery Market Context

Market Characteristics

Singapore possesses highly developed market economies and advanced digital infrastructure, yet **online grocery adoption remains low at 11.9%** of food and beverage purchases (Singstat, 2020).

This paradox highlights the gap between technological capability and consumer behavior

Dominant Players

Traditional supermarket chains maintain market control through extensive physical networks:

NTUC FairPrice	Cold Storage	Sheng Siong
90+ stores	35 supermarkets	Local chain

Emerging Competitors

Pure-play e-commerce firms (Shopee, RedMart/Lazada) are gaining traction with **Shopee capturing the largest market share** among online grocery platforms.

Consumer Behavior

Traditional Preference

Locals prefer physical stores and wet markets due to cultural habits

Tech Adoption

Generation Y (21-29 years) and females show highest online adoption

Household Size

Medium-sized households (2-5 people) more likely to purchase online

Key Market Statistics

Online Grocery Adoption	11.9%
Total Online Consumption	3%
Smartphone Usage	67.7%

Literature Review

Theoretical Framework: Technology Acceptance Model (TAM)

TAM Foundation

Davis, Bagozzi, & Warshaw (1989)

The Technology Acceptance Model was developed to **measure technology acceptance in retail contexts**, providing a robust framework for understanding user adoption of new technologies.

> TAM has become one of the most influential models in technology adoption research

Core Constructs

PU — **Perceived Usefulness**
The degree to which a person believes that using a particular system would enhance their job performance

PEOU — **Perceived Ease of Use**
The degree to which a person believes that using a particular system would be free of effort

→ These constructs are fundamental predictors of technology adoption behavior

Extended Application

This study integrates TAM with additional theoretical perspectives to create a comprehensive framework:

- Convenience context-based concepts
- Emotional perspectives (perceived risk)
- Social influence factors (consumer reviews)

Theoretical Integration

The study combines **cognitive response (TAM)** with **emotional interaction factors** to understand online grocery purchase intention comprehensively.

Integration Models: Theory of Reasoned Action (TRA), convenience context-based concept, and e-WOM literature

Literature Review

Research Variables & Hypotheses

IV1 Consumer Behaviour (CB)

Shopping habits, orientation, and motivation affecting online purchase decisions

Key Elements: Shopping motivation, web design, information quality, payment systems

IV2 Online Shopping Experience (SE)

Prior experience with online tools, shopping habits, and interactive processes

Key Elements: Web atmospherics, prior knowledge, emotional responses

IV3 Convenience (C)

Time-saving, accessibility, search convenience, and transaction ease

Key Elements: Access convenience, transaction ease, possession, post-possession

IV4 Perceived Risk (PR)

Consumer beliefs about potential negative outcomes in online purchasing

Risk Types: Financial, product, delivery, privacy, psychological, time, social

IV5 Consumer Reviews (CR)

Electronic Word-of-Mouth (e-WOM) influencing purchase decisions through trust

Review Types: Relative reviews, anonymous reviews, professional reviews

DV Purchase Intention (PI)

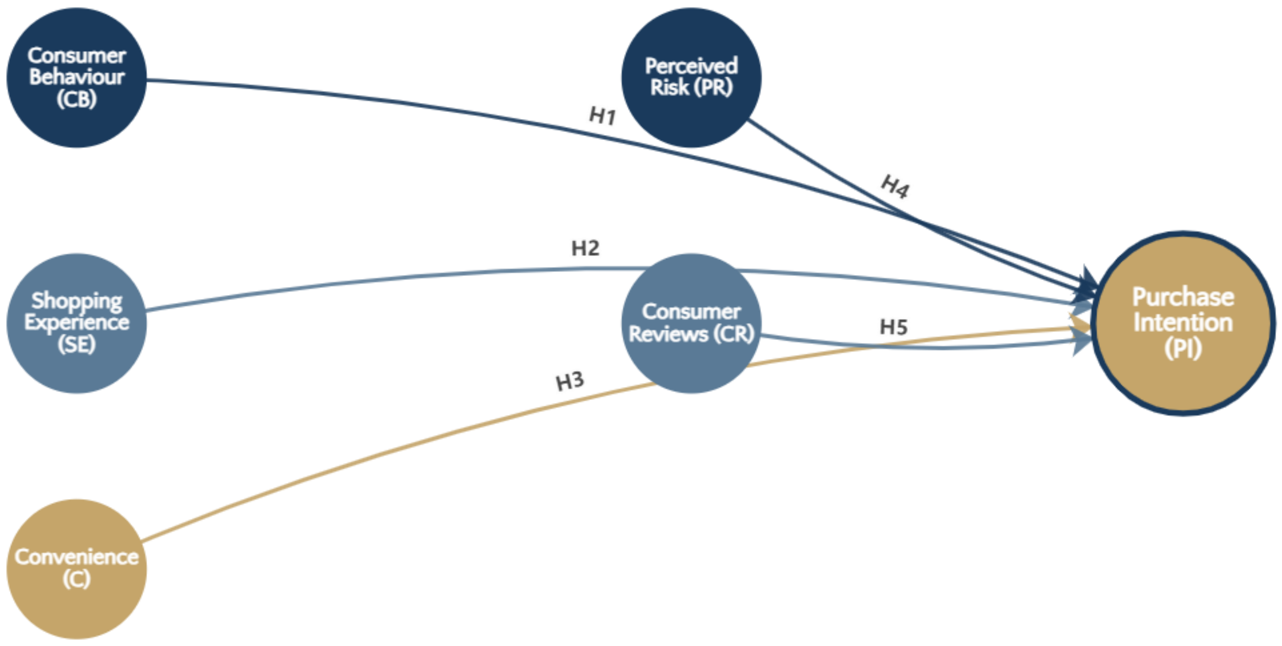
Consumer expectation and possibility of purchasing groceries online

Definition: Expectance of purchasing with reference cost in specific time framework

Conceptual Framework

Conceptual Framework & Hypothesis Mapping

Research Model: Variable Relationships



Five Research Hypotheses

- **H1:** Consumer behaviour → PI (+)
- **H2:** Shopping experience → PI (+)
- **H3:** Convenience → PI (+)
- **H4:** Perceived risk → PI (+)
- **H5:** Consumer reviews → PI (+)

Variable Classification

- **Independent Variables:** CB, SE, C, PR, CR
- **Dependent Variable:** Purchase Intention (PI)

Methodology

Research Methodology

Research Design

Quantitative approach using web-based self-completion survey for hypothesis testing. This method provides:

- Time-effective data collection
- Generalizable findings
- High perceived anonymity
- Reduced social desirability bias

Sampling Strategy

Survey Invitations	150
Valid Responses	129
Response Rate	86%

Distributed via email and social media (Facebook, Twitter, Instagram)

Measurement Scale

Five-point Likert Scale adapted from established literature:

1 = Strongly Disagree — 5 = Strongly Agree

Data Analysis

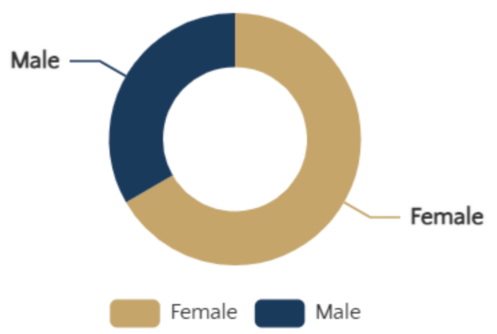
SPSS Software for comprehensive statistical analysis:

- Descriptive Statistics
- Exploratory Factor Analysis (EFA)
- Pearson Correlation Analysis
- Multiple Linear Regression (MLR)

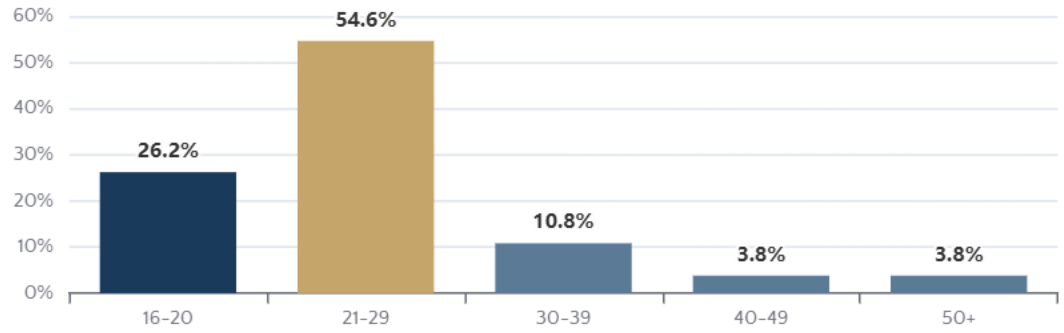
Data Analysis

Sample Demographics & Profile

Gender Distribution (n=129)



Age Composition



Key Demographic Insights

Female Dominance
66.7% of respondents are female, indicating higher online grocery interest among women

Generation Y Focus
54.6% aged 21-29 years, representing tech-savvy consumers

Smartphone Preference
67.7% use smartphones for online purchases

Moderate Engagement
54.6% shop 1-3 times monthly; 63.8% spend <500 SGD

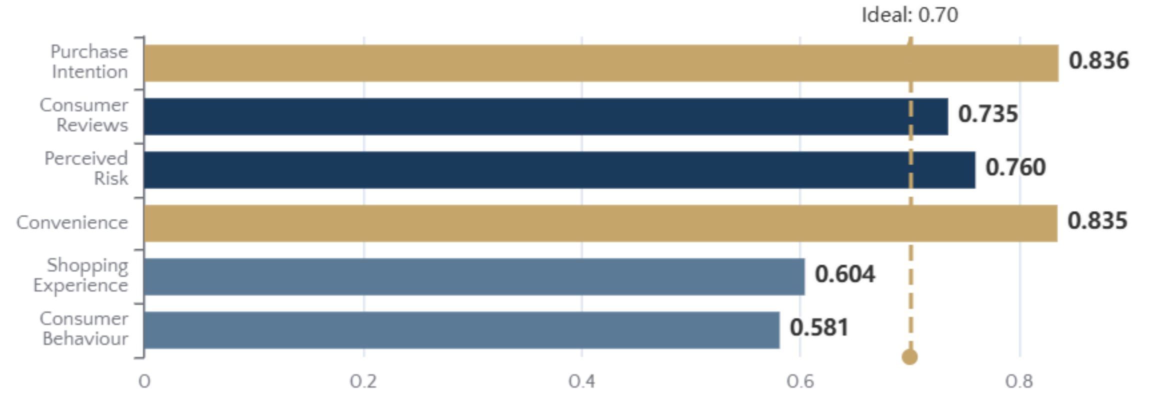
Occupation & Internet Usage

Students	51.5%
Employed	37.7%
>2 Years Internet Use	45.4%

Data Analysis

Reliability Analysis & Factor Validation

Cronbach's Alpha Reliability Results



Acceptable Range: 0.60 - 0.95

Ideal Threshold: > 0.70

Factor Analysis Validity

KMO Measure

0.723

> 0.50 (Acceptable)

Bartlett's Test

$p < 0.001$

Significant

Variance Explained: Component 1 explains 75.389% of total variance, indicating strong construct validity for Purchase Intention

Reliability Assessment

Convenience (C)	0.835
Excellent reliability	
Purchase Intention (PI)	0.836
Excellent reliability	
Perceived Risk (PR)	0.760
Good reliability	
Consumer Reviews (CR)	0.735
Acceptable reliability	

Data Analysis

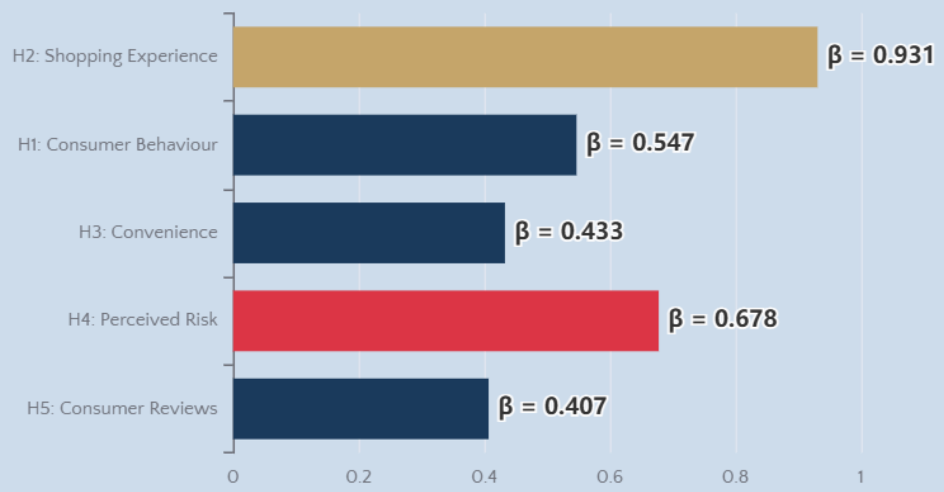
Hypothesis Testing Results: Multiple Linear Regression

Model Fit Statistics

Adjusted R^2	F-Statistic	Durbin-Watson
0.689	57.656	1.952
68.9% variance explained	$p < 0.001$ ***	No autocorrelation

Interpretation: The model explains 68.9% of variance in Purchase Intention, indicating strong predictive power. ANOVA F-test confirms significant relationship ($p < 0.001$).

Path Coefficients Visualization



Hypothesis Testing Summary

Hypothesis	β	p-value	Result
H1: CB → PI	**0.547**	< 0.001***	SUPPORTED
H2: SE → PI	**0.931**	< 0.001***	SUPPORTED
H3: C → PI	**0.433**	< 0.001***	SUPPORTED
H4: PR → PI	**0.678**	0.796 ns	NOT SUPPORTED
H5: CR → PI	**0.407**	< 0.001***	SUPPORTED

*** $p < 0.001$; ns = not significant

Multicollinearity Check

All VIF Values	< 10

Interpretation: VIF values range from 1.074 to 2.460, well below the threshold of 10, indicating no multicollinearity issues.

Key Findings & Discussion

Four Significant Factors

Consumer behaviour, online shopping experience, convenience, and consumer reviews significantly enhance online grocery purchase intention, supporting the extended TAM framework.

Model explains **68.9%** of variance in Purchase Intention

Strongest Predictor

Shopping Experience ($\beta = 0.931$) emerges as the most influential factor. High internet adoption among Generation Y drives positive online purchase intention.

- Positive experience → Higher intention
- Generation Y most responsive

Perceived Risk: Insignificant

H4 not supported ($p = 0.796$), contradicting previous literature. Potential explanations:

- Sample characteristics: tech-savvy respondents less risk-averse
- Singapore's secure e-commerce infrastructure reduces concerns
- Risk affects satisfaction more than initial intention (Ariffin et al., 2018)

Gender & Age Differences

Female Consumers
Show higher susceptibility to influencing factors; dominate personal care and house care purchases

Generation Y (21-29)
Highest online adoption; older generations face technology barriers

Product Category Preferences

Personal Care & House Care	HIGH
Packaged Food	MODERATE
Fresh Food & Produce	LOW

Conclusion

Recommendations & Conclusion

For E-Commerce Firms

1. **Enhance User Experience:** Focus on website design, navigation, and seamless transaction processes
2. **Target Generation Y & Females:** Develop marketing strategies tailored to these high-adoption segments
3. **Engage Older Demographics:** Create innovative, user-friendly approaches to overcome technology barriers

For E-Retailers

1. **Provide Comprehensive Information:** Detailed product descriptions and quality assurance
2. **Maintain Quality Standards:** Reduce perceived risk through consistent product quality
3. **Actively Manage Reviews:** Respond to consumer feedback and address negative reviews promptly

Research Limitations

- Sample skewed toward young, tech-savvy respondents
- Online survey excludes low technology adoption groups
- Perceived risk insignificance may require larger sample validation
- Cross-sectional design limits causal inference

Future Research Directions

→ Expand sample size and age diversity

→ Incorporate qualitative methods (interviews)

→ Examine specific grocery product categories

→ Investigate perceived risk and satisfaction relationship

Conclusion

This study successfully identifies **four key factors** driving online grocery purchase intention in Singapore, contributing valuable insights to this under-researched market segment and providing practical recommendations for e-commerce stakeholders.